

REPORT

Title

Prolog Program for Student-Teacher-Subject Code Database

Aim

To create a Prolog database containing information about **students, teachers, and subject codes** and retrieve related information using queries.

Problem Statement

Create a Prolog database that stores the relationship between students, teachers, and subject codes. The program should allow the user to find the teacher and subject code associated with a particular student.

Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp3.pl`)

Procedure

1. Open **SWI-Prolog**.
2. Create a new file named `exp3.pl`.
3. Enter the student, teacher, and subject-code facts.
4. Save the file.
5. Load the program using `[exp3] ..`
6. Enter a query for a student.

7. Observe the corresponding teacher and subject code.

Program

```
student(pooja, ravi).  
student(rahul, priya).  
student(anjali, kumar).  
student(karthik, ravi).
```

```
teacher(ravi, cs101).  
teacher(priya, cs102).  
teacher(kumar, cs103).
```

```
student_details(Student, Teacher, SubCode) :-  
    student(Student, Teacher),  
    teacher(Teacher, SubCode).
```

Sample Output

Query:

```
?- [exp3].  
true.
```

```
?- student_details(pooja, Teacher, SubCode).
```

Output:

```
Teacher = ravi,  
SubCode = cs101.
```

Another query:

```
?- student_details(rahul, Teacher, SubCode).
```

Output:

```
Teacher = priya,  
SubCode = cs102.
```

Conclusion

Thus, the **Student-Teacher-Subject Code database** was successfully implemented in Prolog. The program can retrieve the teacher and corresponding subject code for a given student using logical queries.